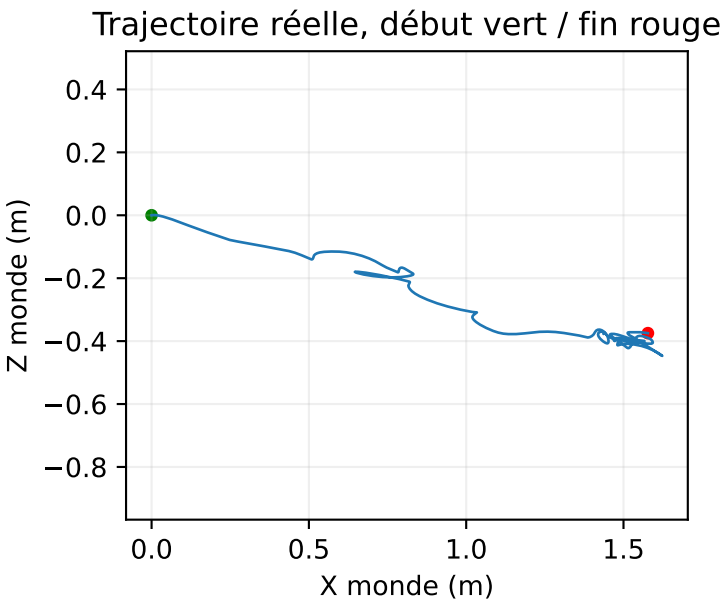
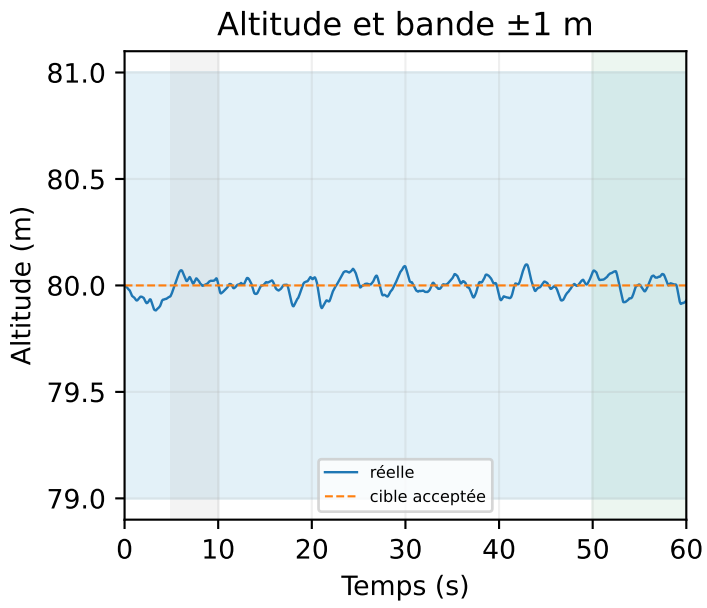
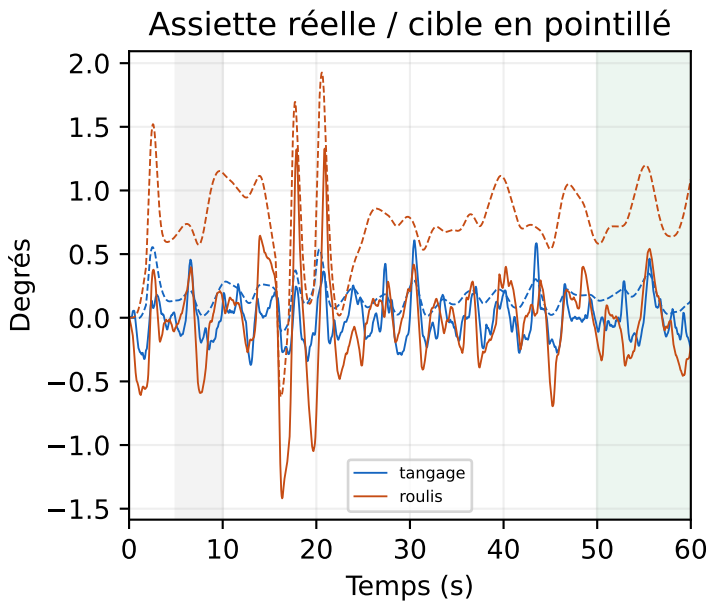
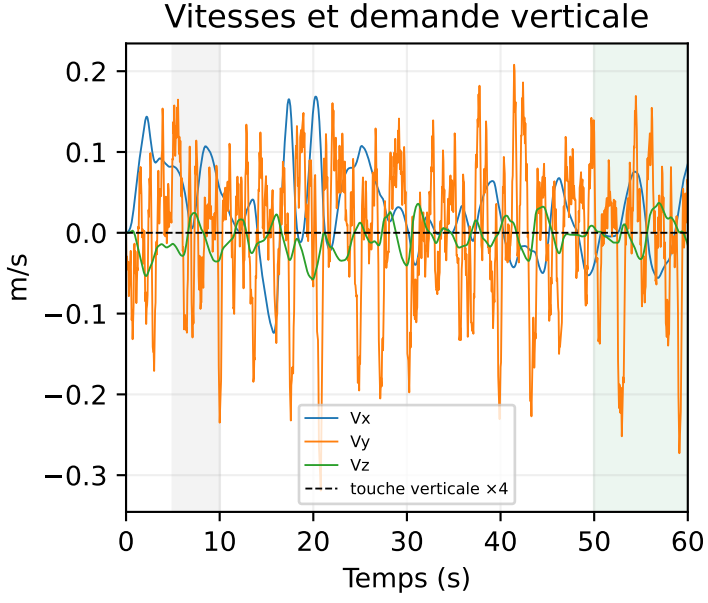
mixed-propellers / fixed / hover — PASS

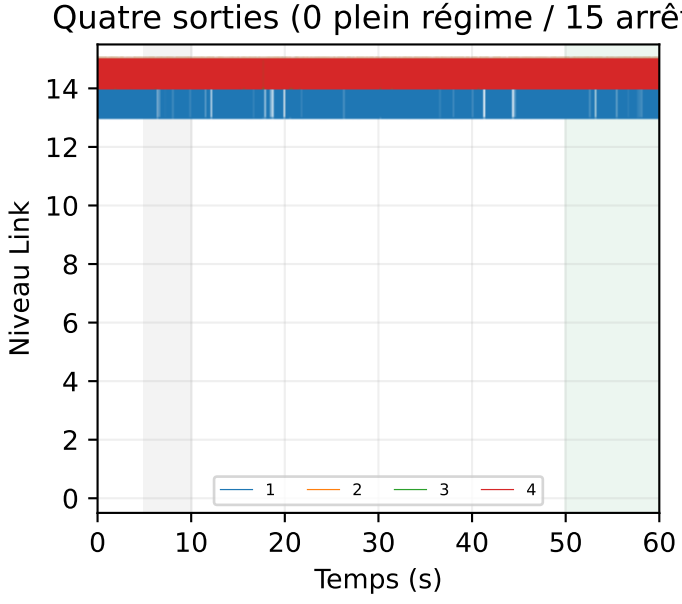
Maintien sans touche pendant toute la simulation.

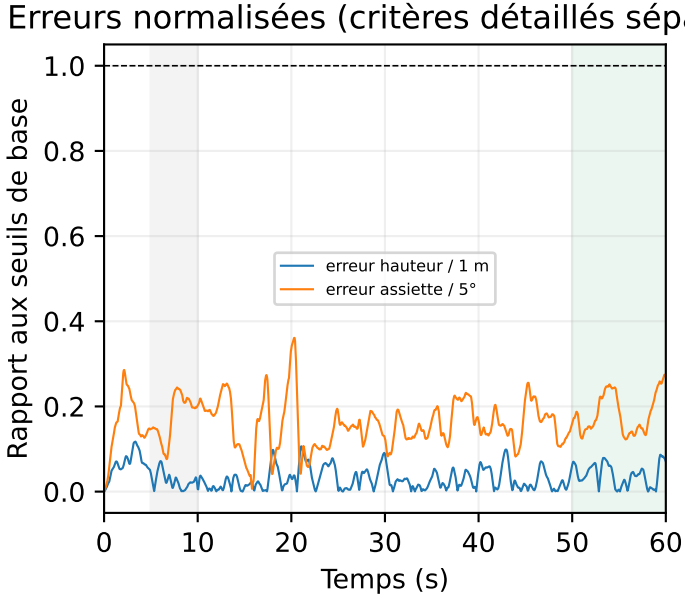










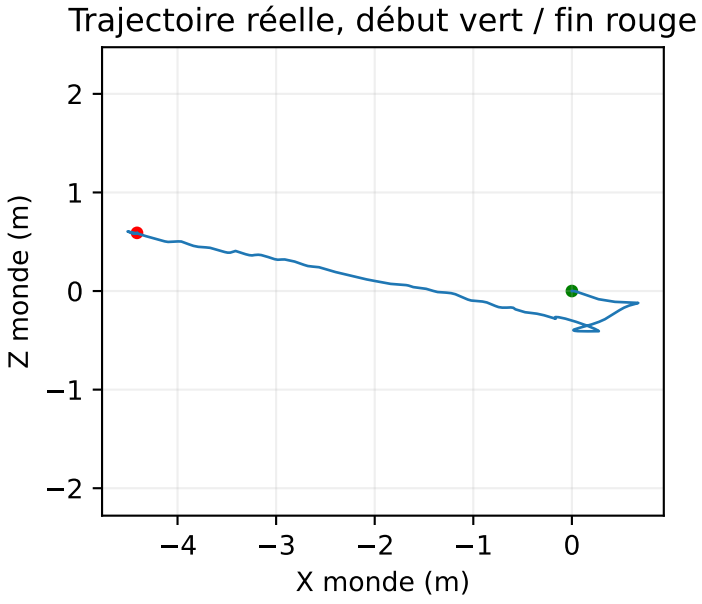


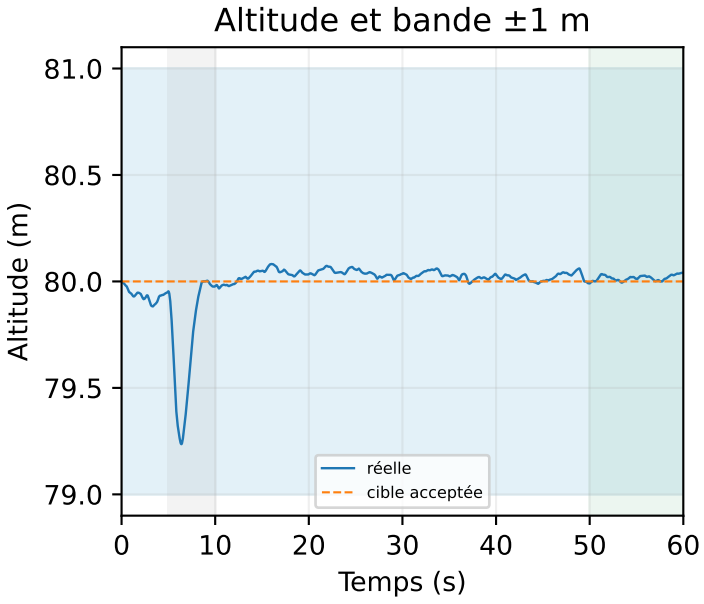
Erreur finale hauteur 0.071 m ; excursion finale 0.157 m ; assiette finale max 0.714° ; freinage — s. Critères échoués : aucun.

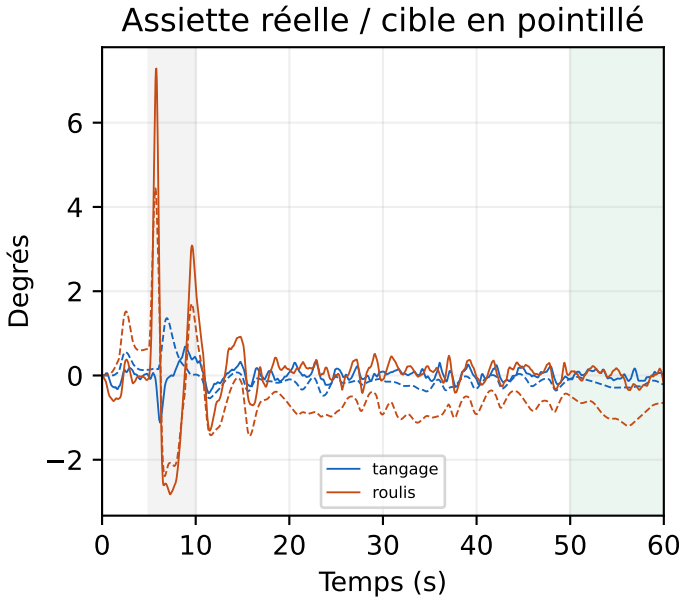
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

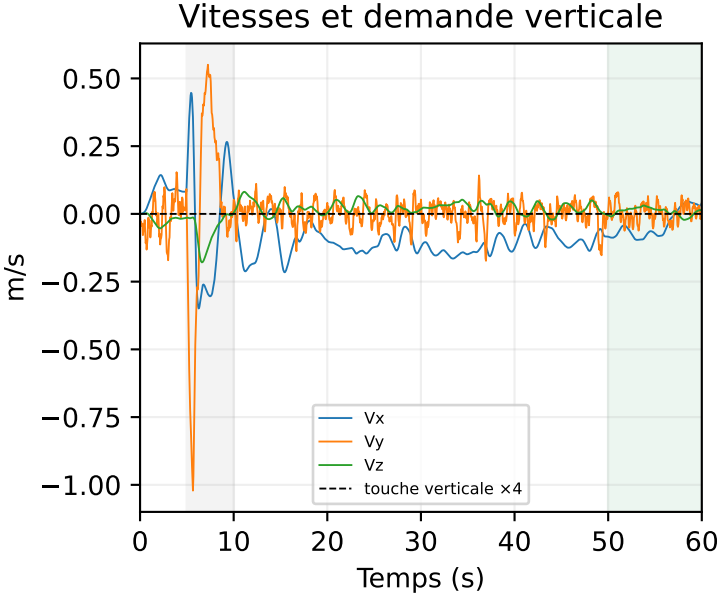
mixed-propellers / fixed / load-swing — PASS

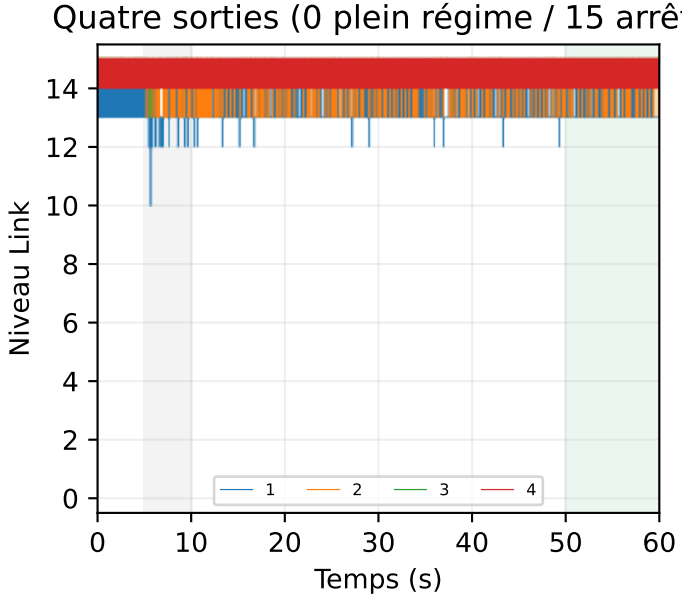
Aucune touche ; charge ponctuelle de 12 kg sur corde 5 m, inclinée initialement de 20°, ajoutée à 5 s.

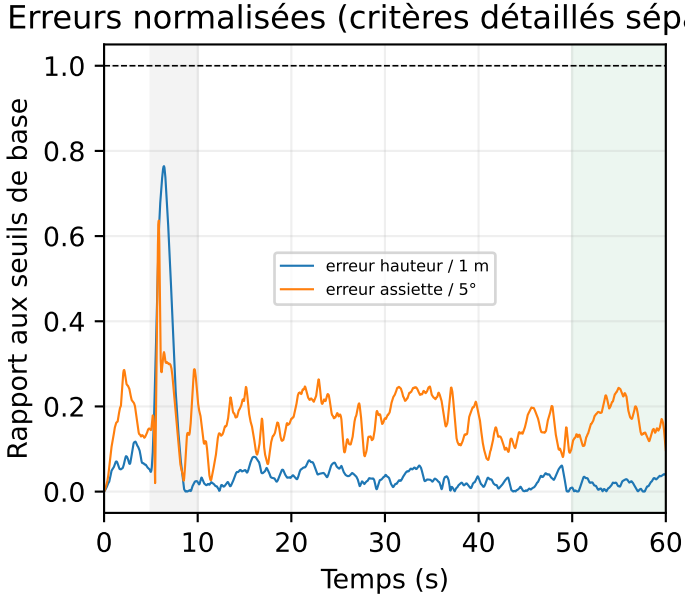










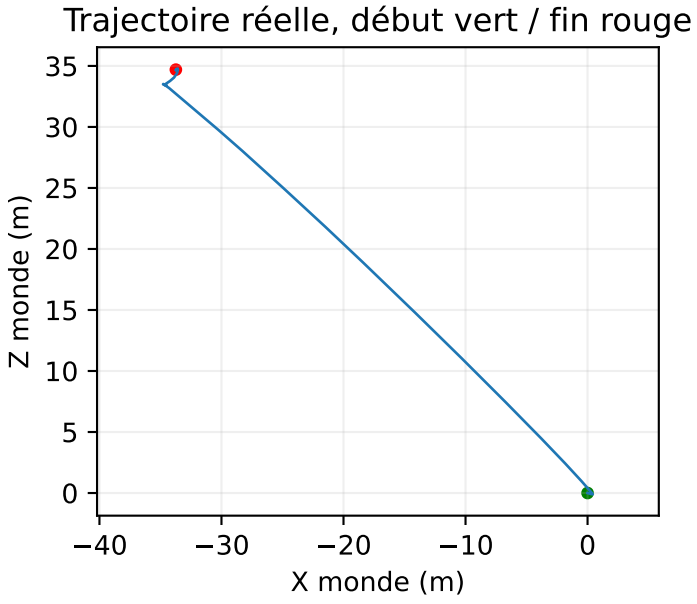


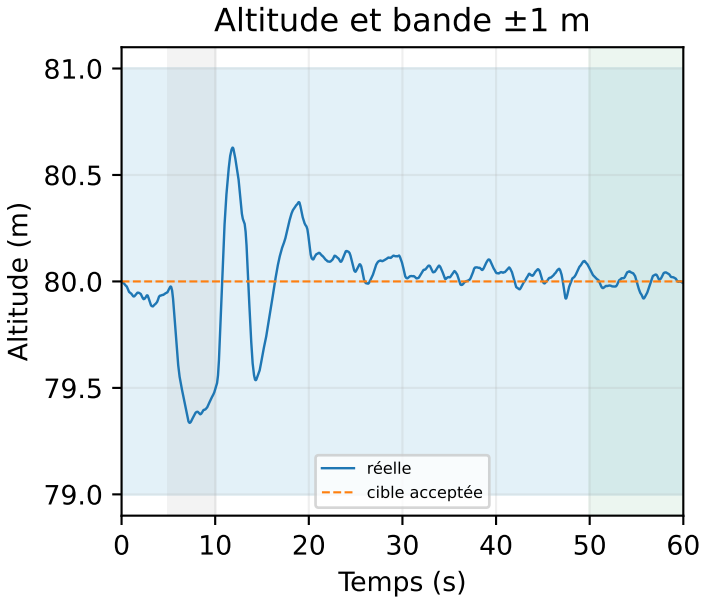
Erreur finale hauteur 0.041 m ; excursion finale 0.049 m ; assiette finale max 0.374° ; freinage — s. Critères échoués : aucun.

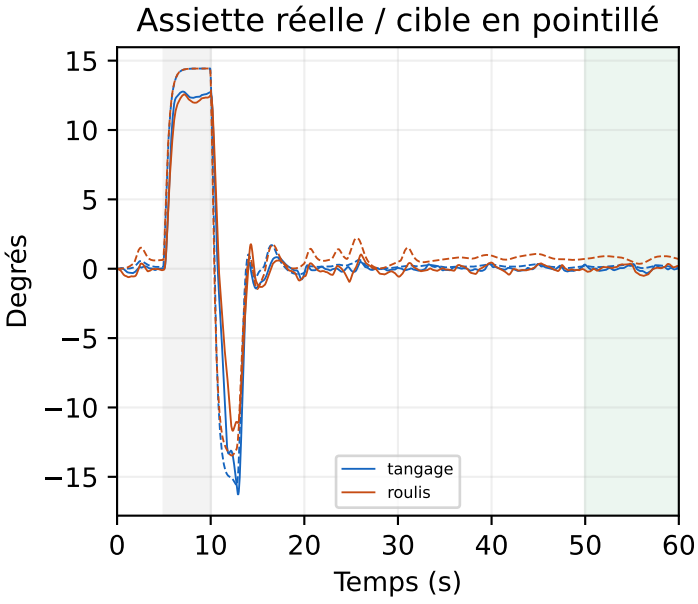
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

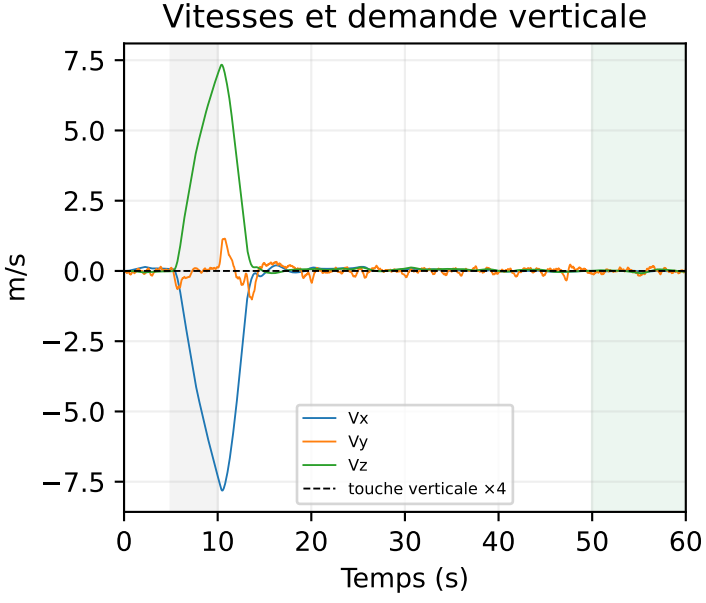
mixed-propellers / fixed / forward-left — PASS

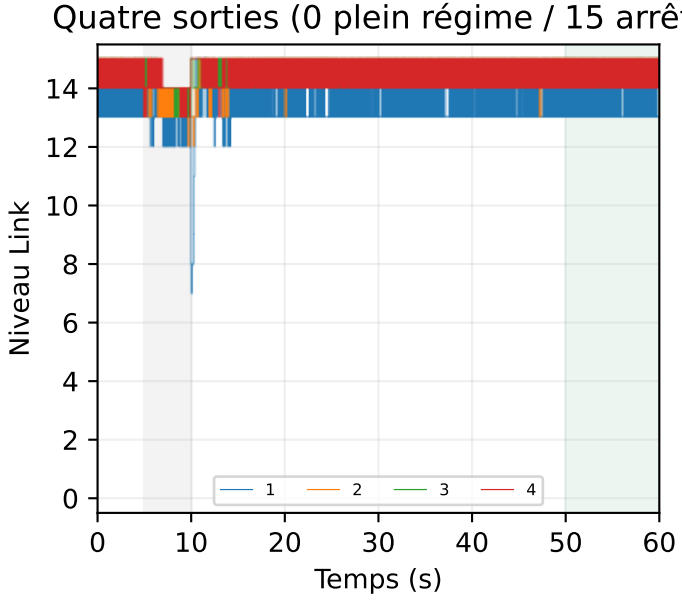
W+Q entre 5 et 10 s, mode déplacement latéral.

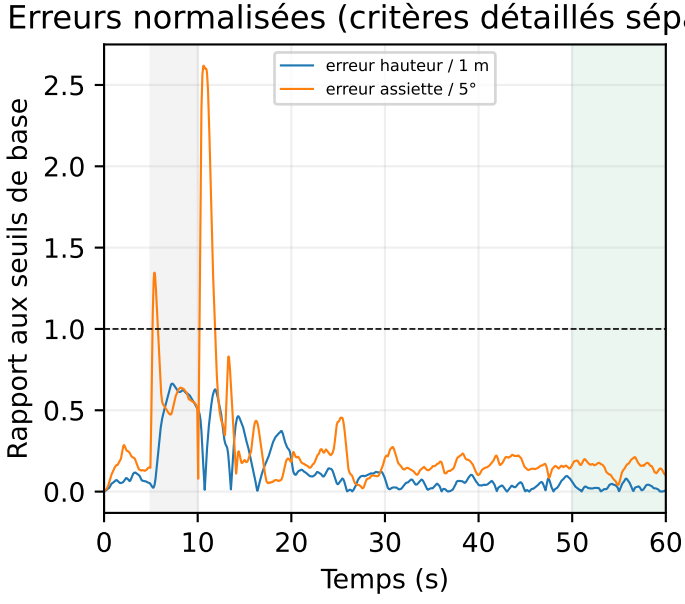










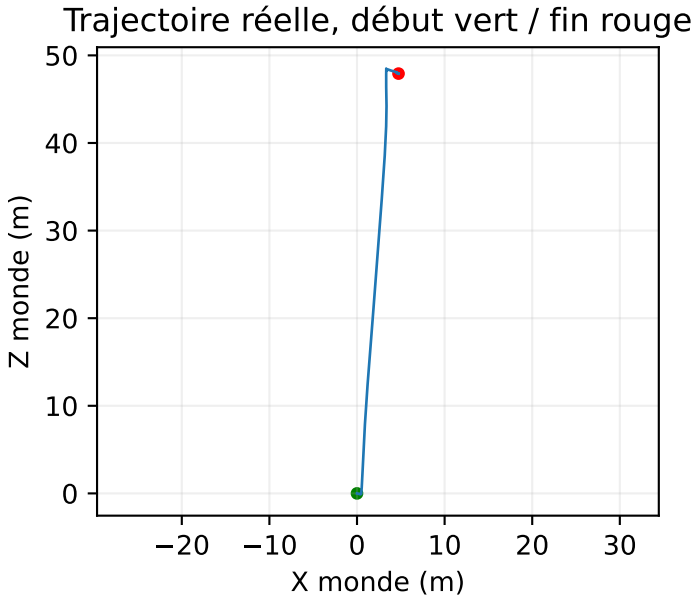


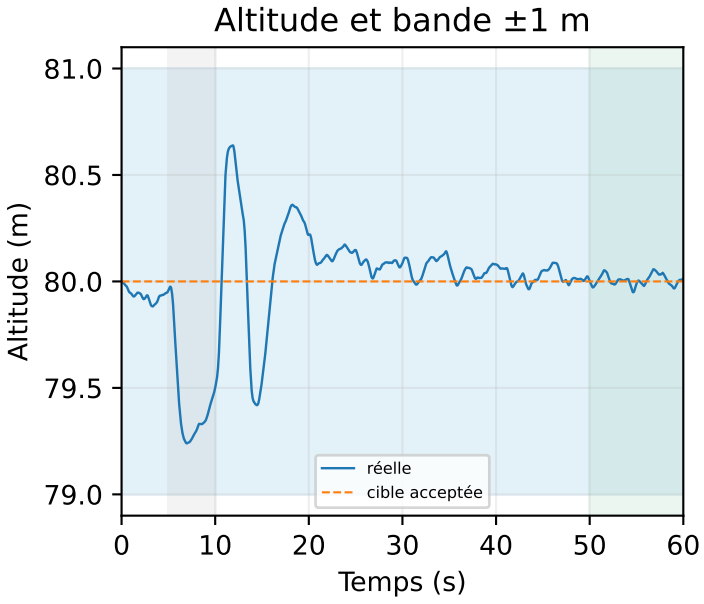
Erreur finale hauteur 0.003 m ; excursion finale 0.136 m ; assiette finale max 0.562° ; freinage 4.400 s. Critères échoués : aucun.

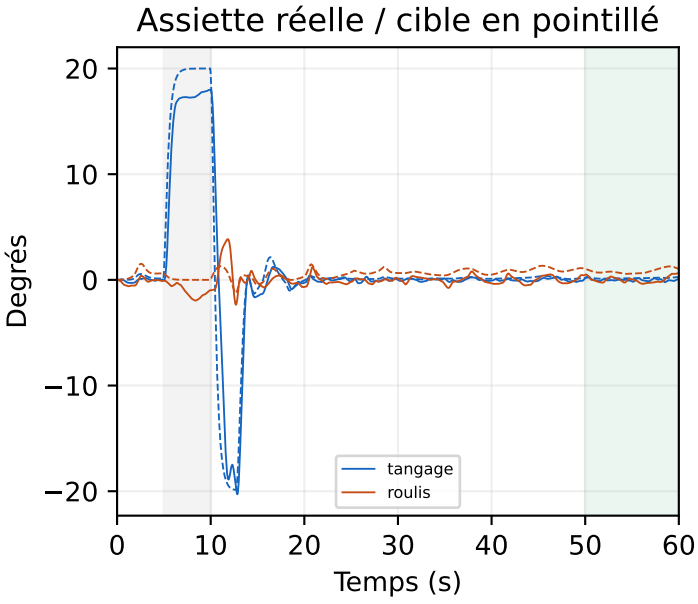
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

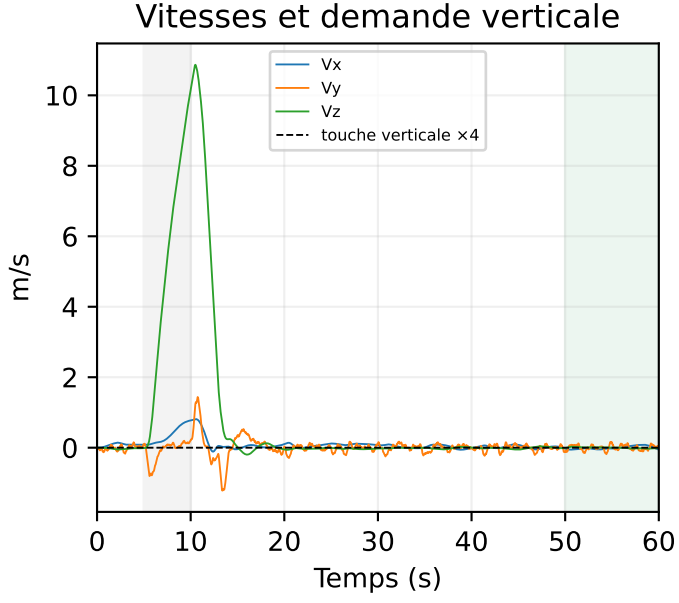
mixed-propellers / fixed / brake — PASS

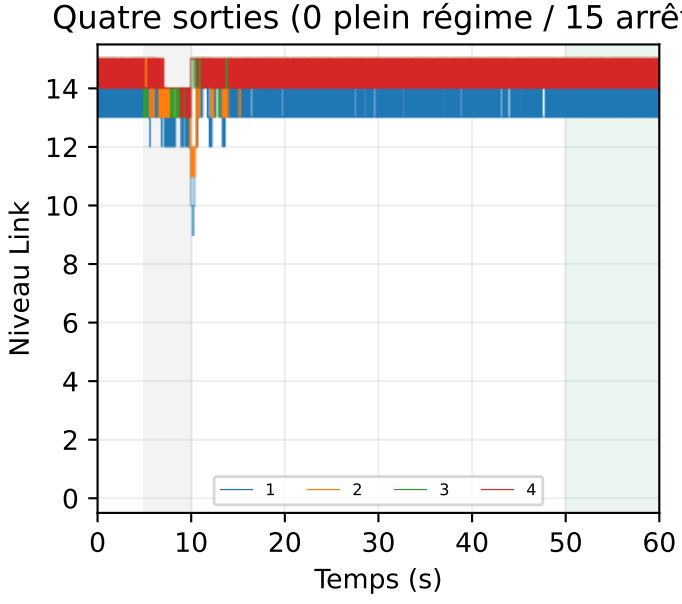
W de 5 à 10 s, relâchement, délai de freinage mesuré sur 15 observations consécutives.

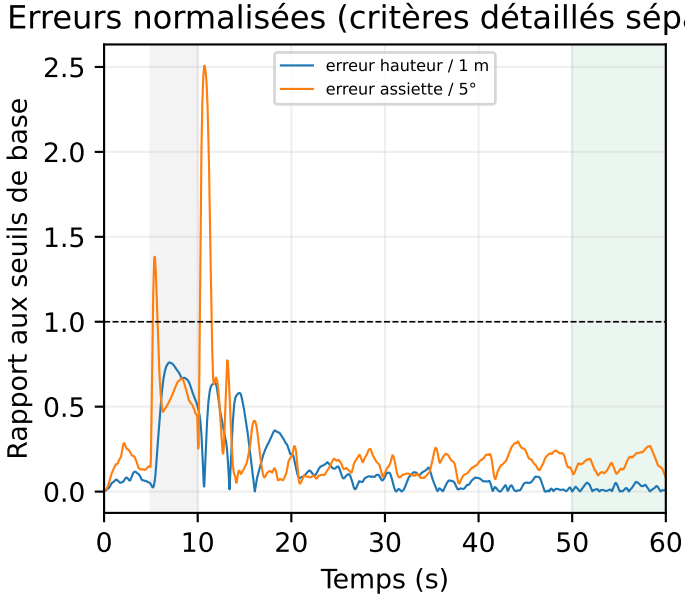










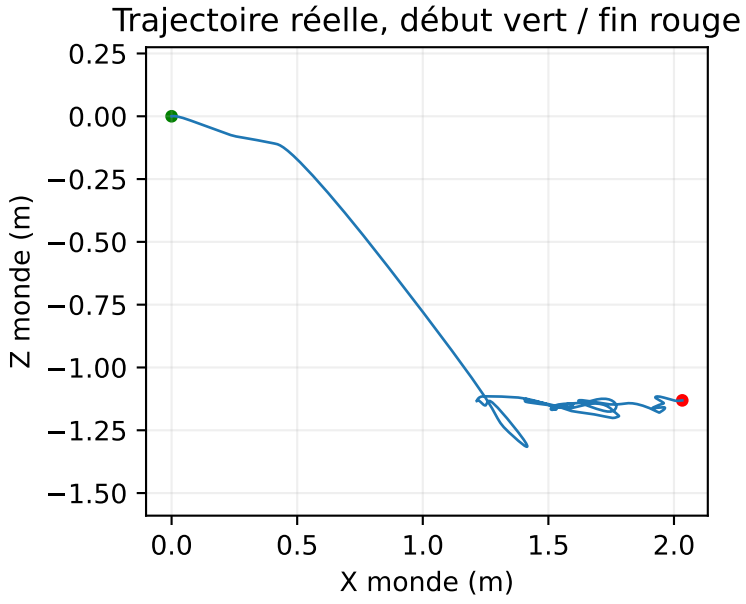


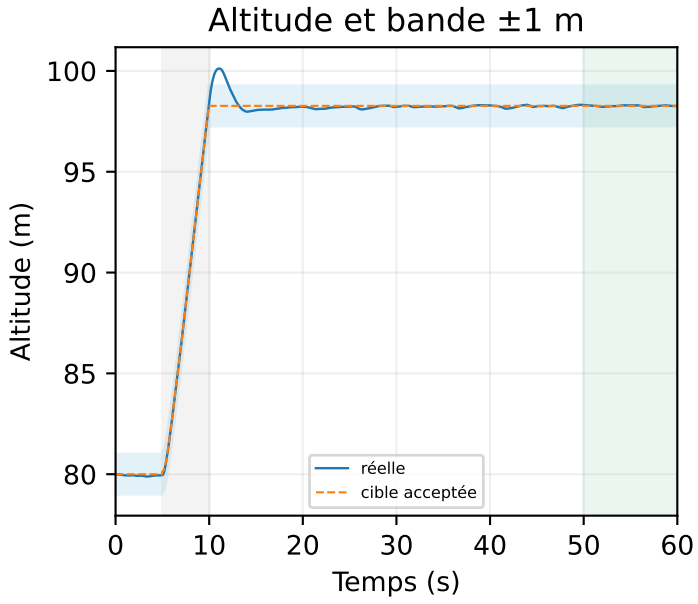
Erreur finale hauteur 0.012 m ; excursion finale 0.109 m ; assiette finale max 0.654° ; freinage 4.400 s. Critères échoués : aucun.

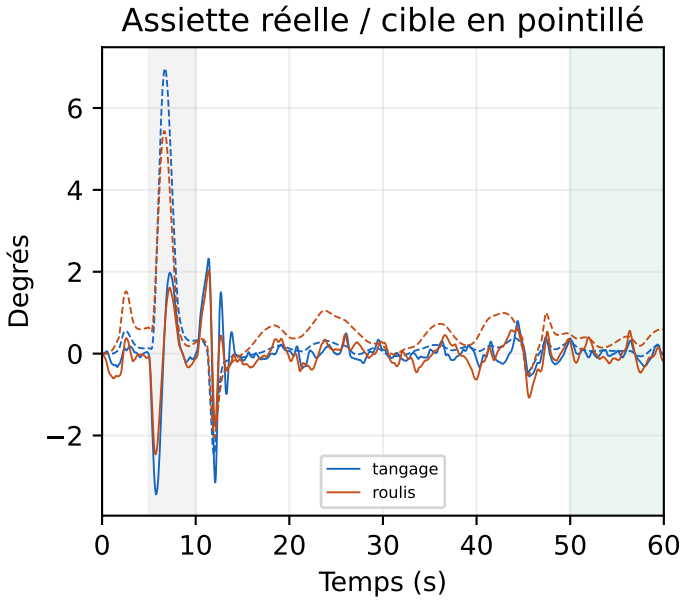
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

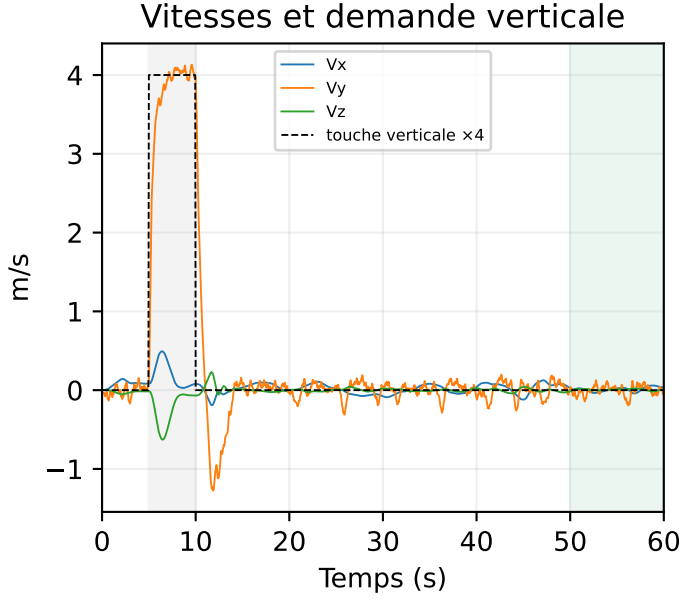
mixed-propellers / fixed / climb — PASS

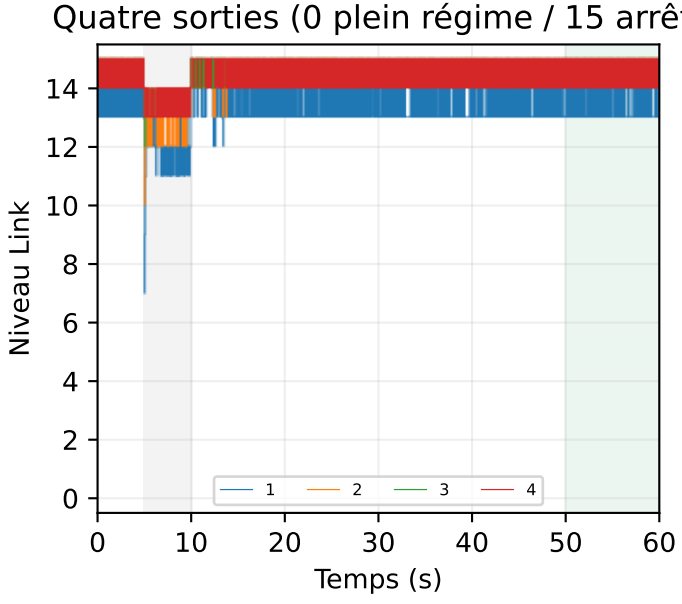
Espace entre 5 et 10 s, puis relâchement et maintien de la hauteur atteinte.

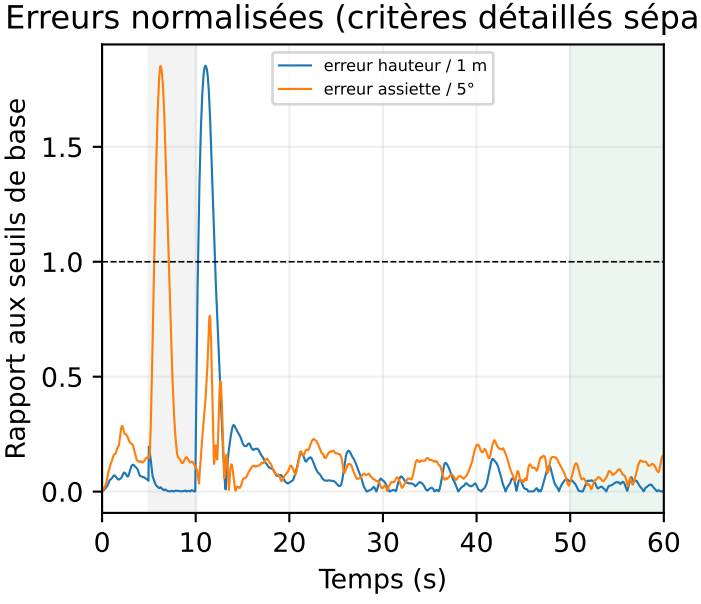










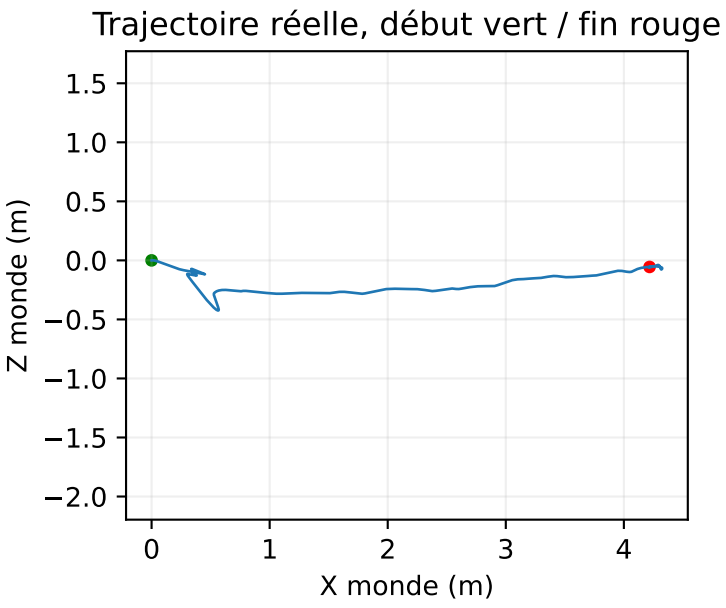


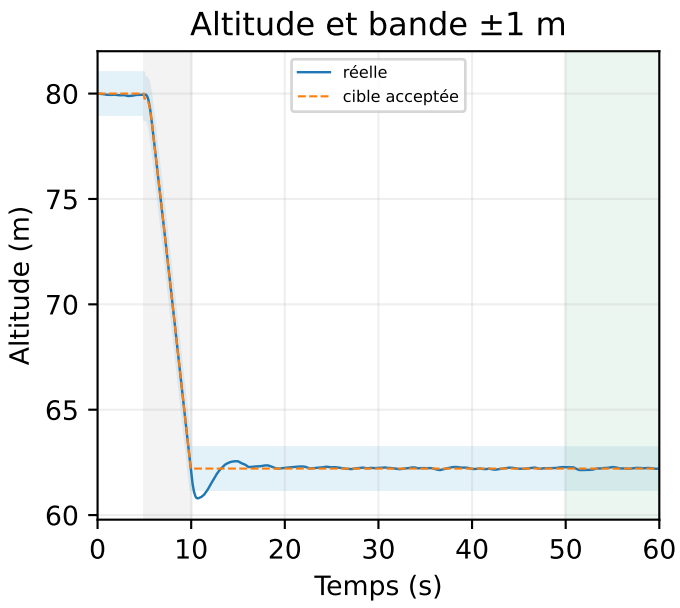
Erreur finale hauteur 0.006 m ; excursion finale 0.112 m ; assiette finale max 0.637° ; freinage — s. Critères échoués : aucun.

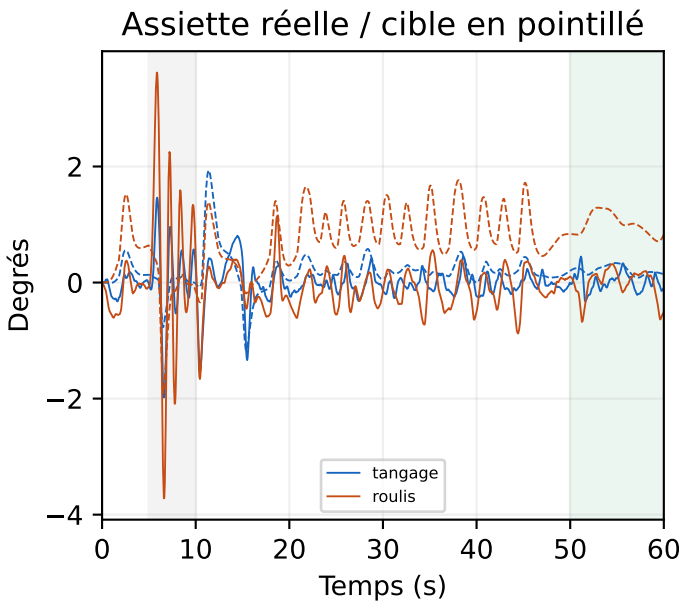
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

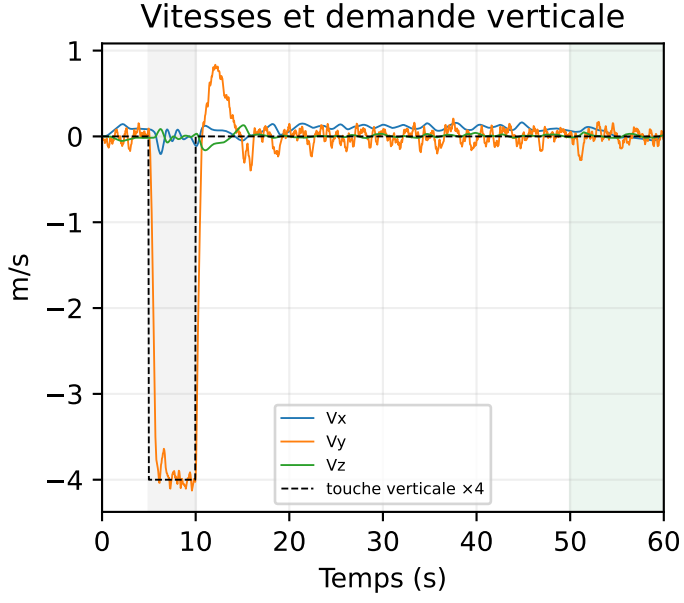
mixed-propellers / fixed / descend — PASS

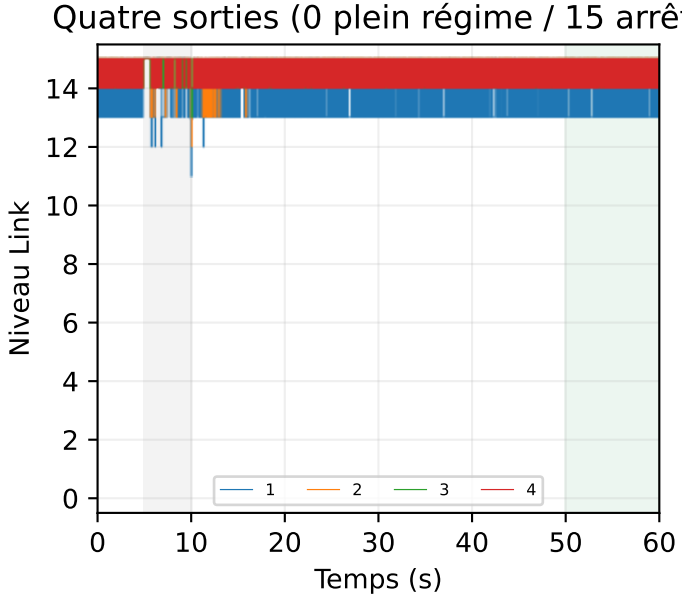
Shift entre 5 et 10 s, puis relâchement et maintien.

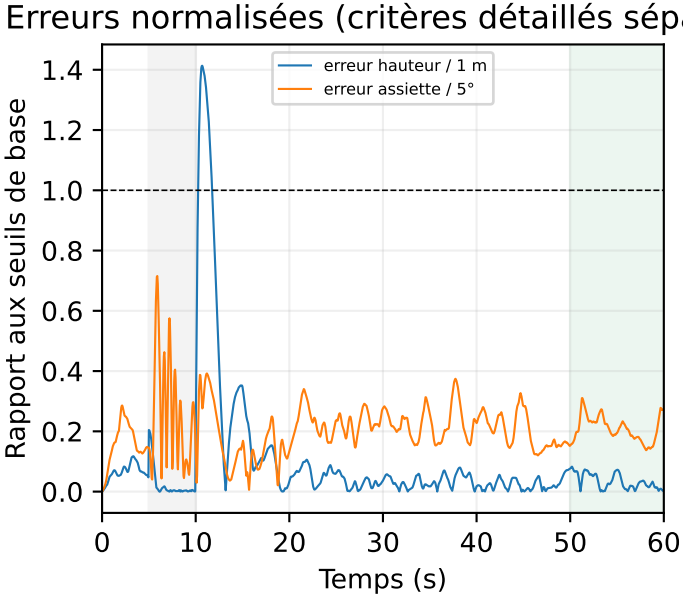










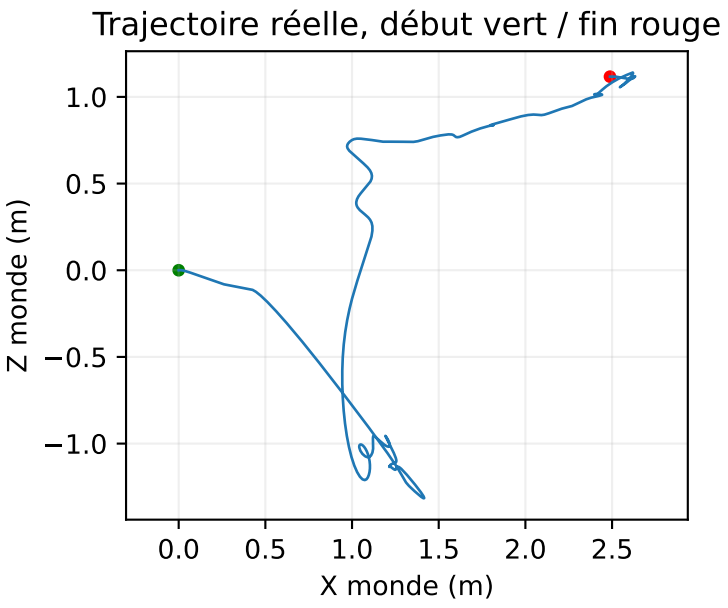


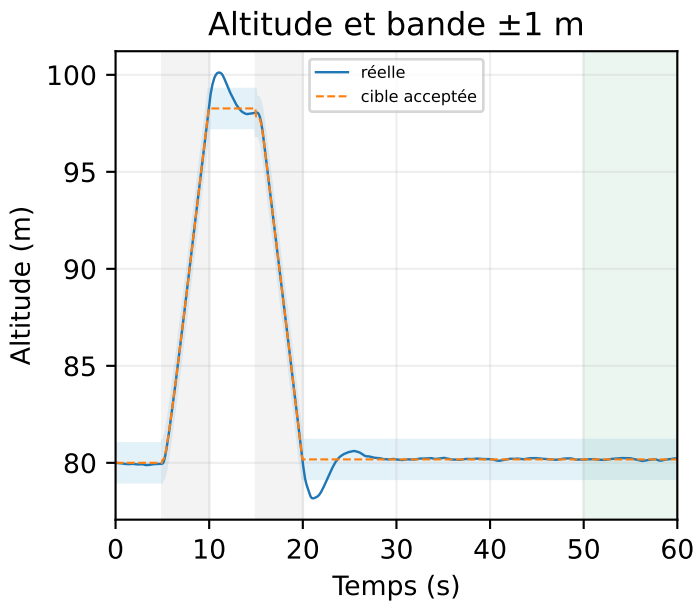
Erreur finale hauteur 0.005 m ; excursion finale 0.156 m ; assiette finale max 0.779° ; freinage — s. Critères échoués : aucun.

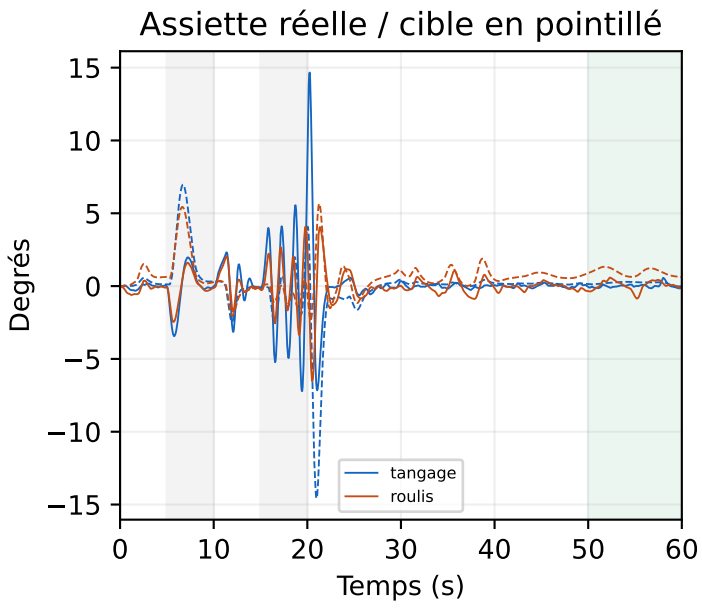
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

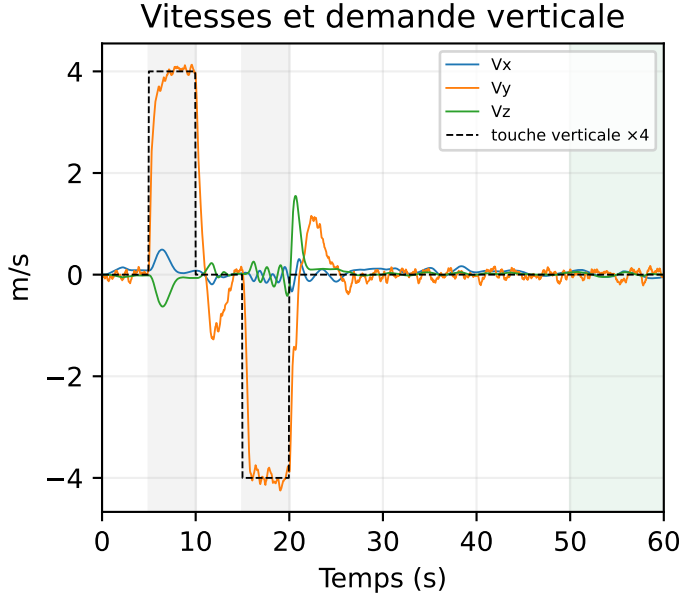
mixed-propellers / fixed / vertical-cycle — FAIL

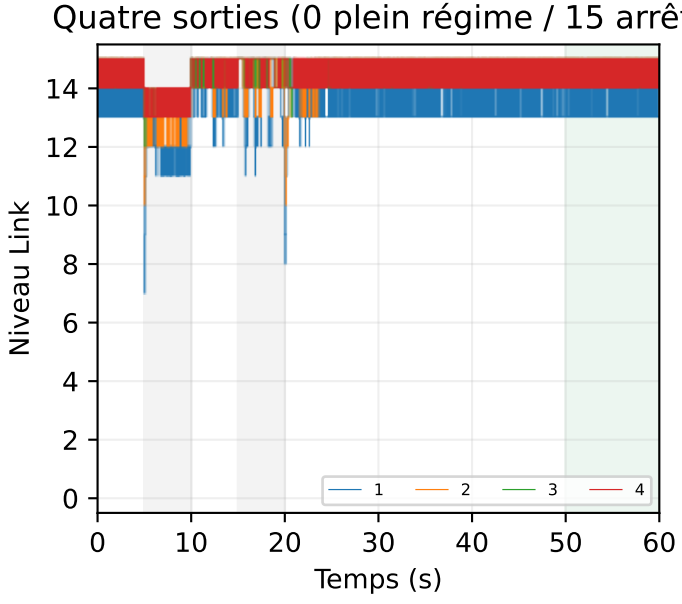
Espace de 5 à 10 s, maintien 10–15 s, Shift 15–20 s, puis maintien.

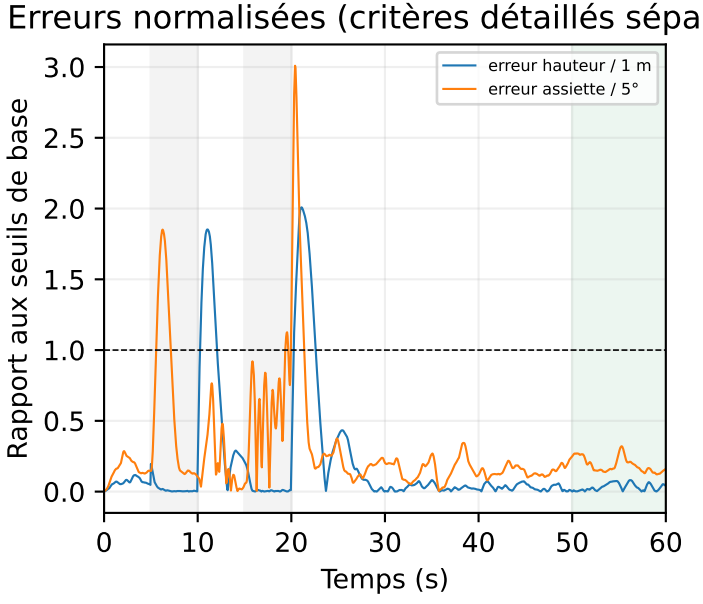












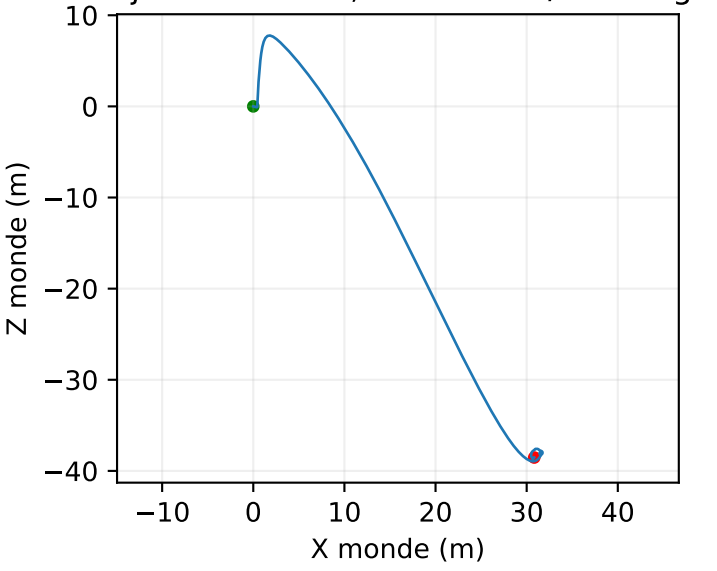
Erreur finale hauteur 0.041 m ; excursion finale 0.156 m ; assiette finale max 0.856° ; freinage — s. Critères échoués : transient_height.

Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

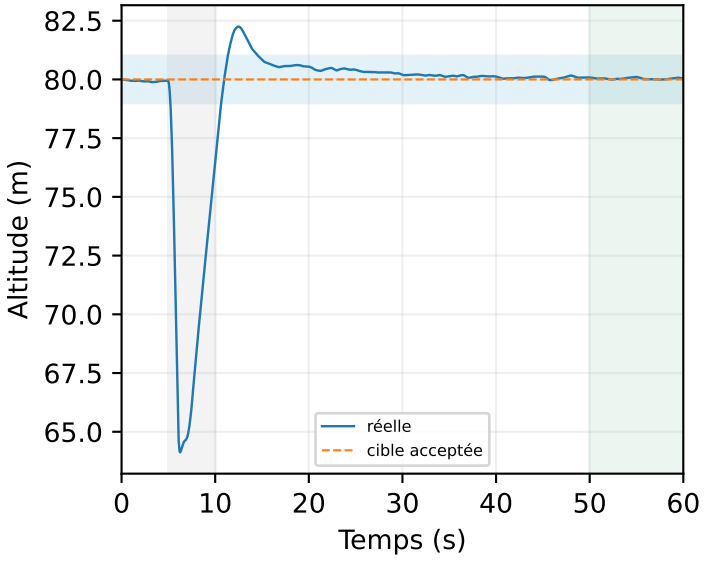
mixed-propellers / fixed / recovery-inverted — FAIL

Aucune touche ; inversion instantanée de 180° à 5 s. Perte d’altitude publiée.

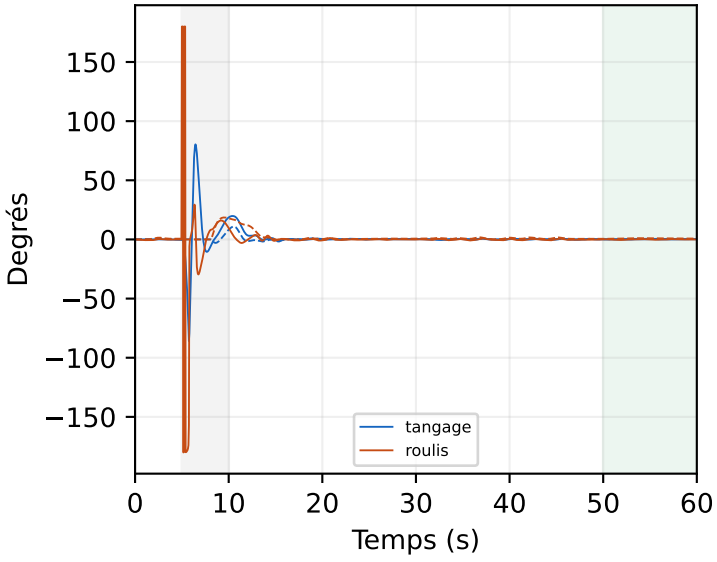
Trajectoire réelle, début vert / fin rouge



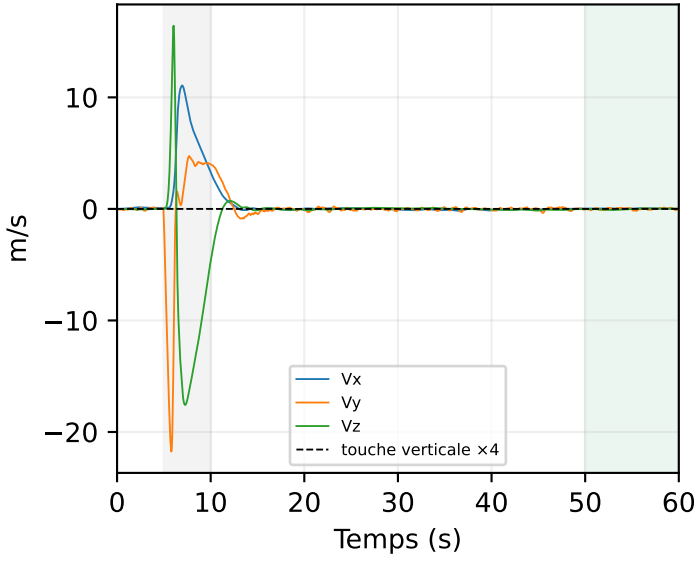
Altitude et bande ±1 m



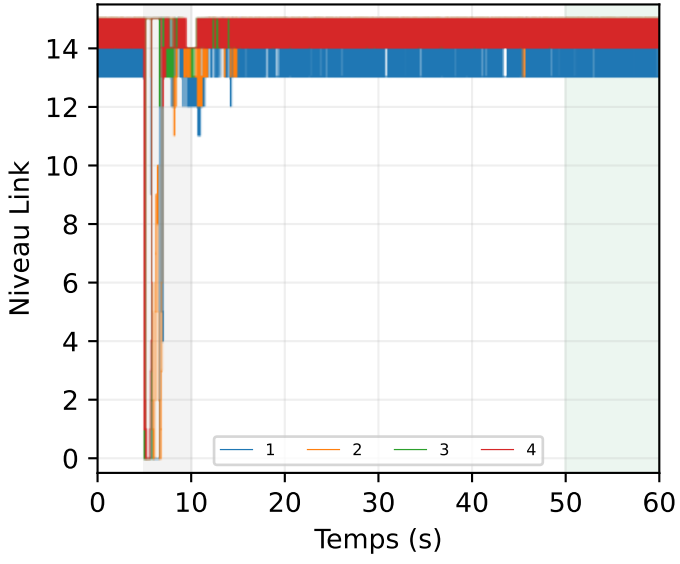
Assiette réelle / cible en pointillé



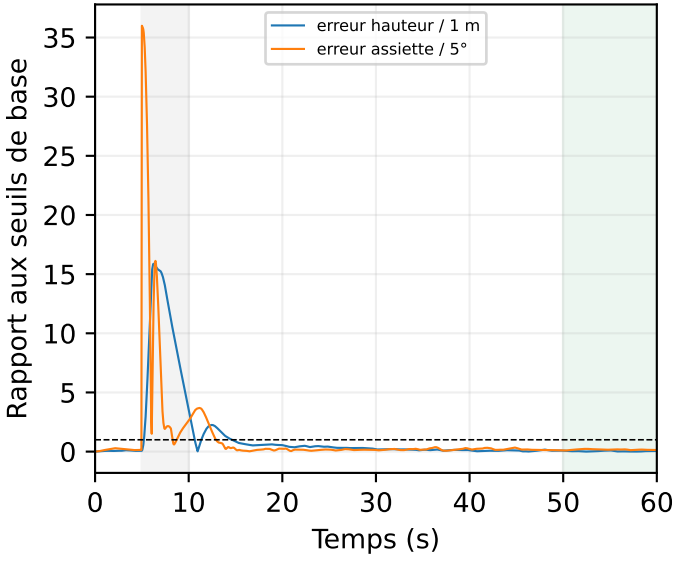
Vitesses et demande verticale



Quatre sorties (0 plein régime / 15 arrêt)



Erreurs normalisées (critères détaillés sépare



Erreur finale hauteur 0.053 m ; excursion finale 0.121 m ; assiette finale max 0.557° ; freinage — s. Critères échoués : transient_height.

Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.